Package 'ssMRCT'

March 28, 2026

Type Package

Title Regional consistency evaluation and sample size calculation for MRCTs

Version 0.1.0

Author Kunhai Qing [cre, aut], Xinru Ren [aut], Jin Xu [aut]

Maintainer Kunhai Qing <kunhaiq@qq.com>

Description A comprehensive suite of functions for designing and analyzing Multi-Regional Clinical Trials (MRCTs) under fixed effects model and random effects model. It features sepcialized tools for consistency probability calculation and optimal regional fraction(s) determination. Support extends two MRCTs (two pivotal, independent MRCTs) under fixed effects model. Developed for regulatory decision support and trial optimization in global drug development programs.

License GPL-3

Encoding UTF-8

LazyData true

NeedsCompilation no

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

Depends R (>= 3.5.0)

Imports cubature, stats, mvtnorm

Contents

ssMRCT-package 2
conProb 3
conProb2 4
conProbII 6
conProbII2 8
CPlbd 10
regATE_dr 12
regFrac 13
regFrac2 15
ssCPbinary 16
ssCPnorm 17
ssCPsurvNPH 19
ssCPsurvPH 22
table2MRCTs 23

Index **25**

`ssMRCT-package` *Regional consistency evaluation and sample size calculation for MRCTs*

Description

The **ssMRCT** package offers a comprehensive suite of functions for designing and analyzing Multi-Regional Clinical Trials (MRCTs), featuring specialized tools for:

- **Consistency Probability Calculation**: Computes the probability that treatment effects across regions satisfy predefined consistency criteria under fixed effects model and random effects model;
- **Optimal Regional Fraction Determination**: Identifies the ideal regional enrollment proportions to minimize regional sample size while maintaining consistency probability under fixed effects model.

Support extends two MRCTs (two pivotal, independent MRCTs) under fixed effects model. Developed for regulatory decision support and trial optimization in global drug development programs. In this package, all sample sizes are rounded up to be integers.

Functions

Key functions included in the package:

conProb, conProb2 Calculate the (conditional) consistency probability under criterion Method I of MHLW (2007) for one/two MRCT(s) with corresponding overall sample size(s) under fixed effects model.

regFrac, regFrac2 Calculate the optimal regional fraction(s) given the (conditional) consistency probability under criterion Method I of MHLW (2007) for one/two MRCT(s), with corresponding overall sample size(s) under fixed effects model.

ssCPnorm, ssCPbinary Calculate the (conditional) consistency probability under criterion Method I of MHLW (2007) for MRCT with corresponding overall sample size under random effects model with normal and binary response, respectively.

ssCPsurvPH, ssCPsurvNPH Calculate the (conditional) consistency probability under criterion Method I of MHLW (2007) for MRCT with corresponding overall sample size under random effects model with survival endpoints (proportional or non-proportional hazard).

References

MHLW (2007). Basic Principles on Global Clinical Trials. https://www.pmda.go.jp/files/000153265.pdf

Kunhai Qing, Xinru Ren, Shuping Jiang, Ping Yang, Menggang Yu and Jin Xu (2025). Regional consistency evaluation and sample size calculation under two MRCTs. Statistics in Medicine 44, no. 25-27 (2025): e70306, https://doi.org/10.1002/sim.70306.

Xinru Ren, Jin Xu (2025). Consistency assessment and regional sample size calculation for MRCTs under random effects model. https://arxiv.org/abs/2508.09443

`conProb` *Consistency probability for MRCT via criterion Method I of MHLW (2007) (conditional version) under fixed effects model*

Description

Calculate the consistency probability for MRCT via criterion Method I of MHLW (2007) (conditional version) under fixed effects model.

Usage

```
conProb(
  alpha,
  power,
  pi = 0.5,
  rF,
  d,
  sigmaTrt,
  sigmaCtrl = sigmaTrt,
  randRatio = 1
)
```

Arguments

`alpha`	The Type I error.
`power`	Power.
`pi`	The threshold ratio in criterion Method I of MHLW (2007) (conditional version). Defaults to 0.5.
`rF`	The regional fraction.
`d`	The true mean of difference of response.
`sigmaTrt`	The standard deviation of response in the treatment group.
`sigmaCtrl`	The standard deviation of response in the control group. Defaults to `sigmaTrt`.
`randRatio`	The randomization ratio between the treatment group and control group. Defaults to 1.

Details

The consistency probability via criterion Method I of MHLW (2007) (conditional version),

$$\Pr\left(D_k \geq \pi D \mid T > z_{1-\alpha}\right),$$

is approximately

$$\frac{1}{1-\beta}\int_{-z_{1-\beta}}^{\infty} \Phi\left(\frac{(1-\pi)(u+z_{1-\alpha}+z_{1-\beta})}{\sqrt{f_k^{-1}-1}}\right)\phi(u)\mathrm{d}u.$$

Since there is no closed form of above equation, `conProb` utilizes the `integrate` function for numerical integration.

As the equation includes none of d, sigmaTrt, sigmaCtrl and randRatio, if only the consistency probability is considered, then the values of d and sigmaTrt could be arbitrary.

The overall sample size is calculated based on the following equation,

$$N^{(c)} = \frac{\left\{r^{-1}\sigma^{2(t)} + \sigma^{2(c)}\right\}(z_{1-\alpha} + z_{1-\beta})^2}{d^2}, \quad N^{(t)} = rN^{(c)}.$$

Then $N = N^{(t)} + N^{(c)}$.

Value

A list containing the following two components:

CP The consistency probability, a scalar.

N The overall sample size.

Examples

```
conProb(alpha = 0.05, power = 0.8, rF = 0.271, d = 1, sigmaTrt = 4)
```

---

conProb2 *Consistency probability for two MRCTs via the extended criterion Method I of MHLW (2007) (conditional version) under fixed effects model*

Description

Calculate the consistency probability for two MRCTs via the extended criterion Method I of MHLW (2007) (conditional version) under fixed effects model.

Usage

```
conProb2(
  alpha,
  power1,
  power2 = power1,
  pi = 0.5,
  rF1,
  rF2 = rF1,
  d1,
  d2 = d1,
  muCtrl1 = 0,
  muCtrl2 = muCtrl1,
  sigmaTrt1,
  sigmaCtrl1 = sigmaTrt1,
  sigmaTrt2 = sigmaTrt1,
  sigmaCtrl2 = sigmaTrt2,
  randRatio1 = 1,
  randRatio2 = randRatio1
)
```

Arguments

alpha	The Type I error.
power1	Power for MRCT 1.
power2	Power for MRCT 2. Defaults to power1.
pi	The threshold ratio in the extended criterion Method I of MHLW (2007) (conditional version). Defaults to 0.5.
rF1	The regional fraction for MRCT 1.
rF2	The regional fraction for MRCT 2. Defaults to rF1.
d1	The true mean of difference of response for MRCT 1.
d2	The true mean of difference of response for MRCT 2. Defaults to d1.
muCtrl1	The true mean of response in the control group for MRCT 1. Defaults to 0.
muCtrl2	The true mean of response in the control group for MRCT 2. Defaults to muCtrl1.
sigmaTrt1	The standard deviation of response in the treatment group for MRCT 1.
sigmaCtrl1	The standard deviation of response in the control group for MRCT 1. Defaults to sigmaTrt1.
sigmaTrt2	The standard deviation of response in the treatment group for MRCT 2. Defaults to sigmaTrt1.
sigmaCtrl2	The standard deviation of response in the control group for MRCT 2. Defaults to sigmaTrt2.
randRatio1	The randomization ratio between the treatment group and control group for MRCT 1. Defaults to 1.
randRatio2	The randomization ratio between the treatment group and control group for MRCT 2. Defaults to randRatio1.

Details

The extended consistency probability via the extended criterion Method I of MHLW (2007) (conditional version),

$$\Pr\left(D_{k,\text{pool}} \geq \pi D_{\text{pool}} \mid T^{(1)} > z_{1-\alpha}, T^{(2)} > z_{1-\alpha}\right),$$

is approximately

$$\frac{1}{(1-\beta_1)(1-\beta_2)} \int_{-z_{1-\beta_1}}^{\infty} \int_{-z_{1-\beta_2}}^{\infty} \Phi\left(\frac{\mu_{L|D}(u,v)}{\sigma_{L|D}}\right) \phi(u)\phi(v)\mathrm{d}u\mathrm{d}v,$$

where the conditional mean $\mu_{L|D}$ and conditional variance $\sigma_{L|D}$ are given in Qing et al (2025).

Since there is no closed forms of above equations, conProb2 utilizes the integrate function for numerical integration.

Additionally, if we assume equal treatment effects, common regional fractions, common powers and common randomization ratios across two studies, i.e., $\sigma^{2(\mathrm{t},1)} = \sigma^{2(\mathrm{t},2)}, \sigma^{2(\mathrm{c},1)} = \sigma^{2(\mathrm{c},2)}, d^{(1)} = d^{(2)}, f_k^{(1)} = f_k^{(2)} = f_k, \beta_1 = \beta_2$ and $r^{(1)} = r^{(2)}$, then the above equation reduces to

$$\frac{1}{(1-\beta)^2} \int_{-z_{1-\beta}}^{\infty} \int_{-z_{1-\beta}}^{\infty} \Phi\left(\frac{(1-\pi)(2z_{1-\alpha} + 2z_{1-\beta} + u + v)}{\sqrt{2(f_k^{-1} - 1)}}\right) \phi(u)\phi(v)\mathrm{d}u\mathrm{d}v.$$

Since now the above equation includes none of d1=d2, sigmaTrt1=sigmaTrt2, sigmaCtrl1=sigmaCtrl2 randRatio1 = randRatio2, if only the consistency probability is considered, then the values of d1 and sigmaTrt1 could be arbitrary.

The overall sample size is calculated in the same way as conProb.

Value

A list containing the following two components:

`CP` The consistency probability, a scalar.

`N1` The overall sample size for MRCT 1.

`N2` The overall sample size for MRCT 2.

Examples

```
conProb2(alpha = 0.05, power1 = 0.8, power2 = 0.9, rF1 = 0.141, d1 = 1, sigmaTrt1 = 4)
```

`conProbII` *Consistency probability for MRCT via criterion Method II of MHLW (2007) (conditional version) under fixed effects model*

Description

Calculate the consistency probability for MRCT via criterion Method II of MHLW (2007) (conditional version) under fixed effects model.

Usage

```
conProbII(
  alpha,
  power,
  rF,
  d,
  sigmaTrt,
  sigmaCtrl = sigmaTrt,
  pTrt,
  pCtrl,
  randRatio = 1,
  responseType = c("continuous", "binary"),
  B = 1e+05
)
```

Arguments

`alpha`	The Type I error.
`power`	Power.
`rF`	The regional fractions, a `K x 1` vector with each component representing the regional fraction in region k and the sum must equal 1.
`d`	The true mean of difference of response.
`sigmaTrt`	The standard deviation of response in the treatment group.
`sigmaCtrl`	The standard deviation of response in the control group. Defaults to `sigmaTrt`.
`pTrt`	The mean of the response in the treatment group.
`pCtrl`	The mean of the response in the control group.

randRatio	The randomization ratio between the treatment group and control group. Defaults to 1.
responseType	The type of response. One of "continuous" and "binary".
B	The number of simulation by Monto Carlo for responseType = "binary". Defaults to 100,000.

Details

The consistency probability via criterion Method II of MHLW (2007) (conditional version),

$$\Pr\left(D_k \geq 0, k = 1, \ldots, K \mid T > z_{1-\alpha}\right),$$

is approximately

$$\frac{1}{1-\beta} \int_{-z_{1-\beta}}^{\infty} \Phi_K \left\{\mathbf{0}; -(u + z_{1-\alpha} + z_{1-\beta})\mathbf{1}_K, \mathbf{\Lambda} - \mathbf{1}_K \mathbf{1}_K^\top\right\} \phi(u) \mathrm{d}u,$$

where $\Phi_K(\mathbf{x}; \vec{\mu}, \Sigma)$ denotes the cumulative distribution function of a K-dimensional normal distribution with mean $\vec{\mu}$ and covariance matrix Σ, $\mathbf{1}_K$ is the vector of K ones, $\mathbf{\Lambda} = \mathrm{diag}(f_1^{-1}, \ldots, f_K^{-1})$.

Since there is no closed form of above equation, conProbII utilizes the pmvnorm function for numerical integration.

As the first equation includes none of d, sigmaTrt, sigmaCtrl and randRatio, if only the consistency probability is considered, then the values of d and sigmaTrt could be arbitrary.

For binary response, the above approximation loses precision under moderate sample size. Hence conProbII applies Monto Carlo to calculate the correct consistency probability.

The overall sample size is calculated in the same way as conProb.

Value

A list containing the following two components:

CP The consistency probability, a scalar.

N The overall sample size.

Examples

```
# Example 1 in Qing et al (2025)
alpha <- 0.05
power <- 0.8
conProbII(alpha, power, rF = rep(1/2,2), d = 1, sigmaTrt = 4, responseType = "continuous")
conProbII(alpha, power, rF = rep(1/3,3), d = 1, sigmaTrt = 4, responseType = "continuous")
conProbII(alpha, power, rF = rep(1/4,4), d = 1, sigmaTrt = 4, responseType = "continuous")
rFk1 <- 0.110
rFk2 <- rFk3 <- (1-rFk1)/2
rF <- c(rFk1, rFk2, rFk3)
conProbII(alpha, power, rF = rF, d = 1, sigmaTrt = 4, responseType = "continuous")
# Example 2 in Qing et al (2025)
rFk1 <- 0.149
rFk2 <- rFk3 <- (1-rFk1)/2
rF <- c(rFk1, rFk2, rFk3)
set.seed(123)
conProbII(alpha, power, rF = rF, pTrt = 0.8, pCtrl = 0.7, responseType = "binary")
rFk1 <- 0.14
rFk2 <- rFk3 <- (1-rFk1)/2
```

```
rF <- c(rFk1, rFk2, rFk3)
set.seed(123)
conProbII(alpha, power, rF = rF, pTrt = 0.7, pCtrl = 0.6, responseType = "binary")
rFk1 <- 0.110
rFk2 <- rFk3 <- (1-rFk1)/2
rF <- c(rFk1, rFk2, rFk3)
set.seed(123)
conProbII(alpha, power, rF = rF, pTrt = 0.8, pCtrl = 0.7, responseType = "binary")
conProbII(alpha, power, rF = rF, pTrt = 0.7, pCtrl = 0.6, responseType = "binary")
```

`conProbII2` *Consistency probability for two MRCTs via criterion Method II (conditional version)*

Description

Calculate the consistency probability for two MRCTs via criterion Method II of MHLW (2007) (conditional version) under fixed effects model.

Usage

```
conProbII2(
  alpha,
  power1,
  power2 = power1,
  rF1,
  rF2 = rF1,
  d1,
  d2 = d1,
  sigmaTrt1,
  sigmaCtrl1 = sigmaTrt1,
  sigmaTrt2 = sigmaTrt1,
  sigmaCtrl2 = sigmaTrt2,
  pTrt1,
  pCtrl1,
  pTrt2 = pTrt1,
  pCtrl2 = pCtrl1,
  randRatio1 = 1,
  randRatio2 = randRatio1,
  responseType = c("continuous", "binary"),
  B = 1e+05
)
```

Arguments

`alpha`	The Type I error.
`power1`	Power for MRCT 1.
`power2`	Power for MRCT 2. Defaults to `power1`.
`rF1`	The regional fractions for MRCT 1, a `K x 1` vector with each component representing the regional fraction in region k and the sum must equal 1.

rF2	The regional fractions for MRCT 2, a K x 1 vector with each component representing the regional fraction in region k and the sum must equal 1. Defaults to rF1.
d1	The true mean of difference of response for MRCT 1.
d2	The true mean of difference of response for MRCT 2. Defaults to d1.
sigmaTrt1	The standard deviation of response in the treatment group for MRCT 1.
sigmaCtrl1	The standard deviation of response in the control group for MRCT 1. Defaults to sigmaTrt1.
sigmaTrt2	The standard deviation of response in the treatment group for MRCT 2. Defaults to sigmaTrt1.
sigmaCtrl2	The standard deviation of response in the control group for MRCT 2. Defaults to sigmaTrt2.
pTrt1	The mean of the response in the treatment group for MRCT 1.
pCtrl1	The mean of the response in the control group for MRCT 1.
pTrt2	The mean of the response in the treatment group for MRCT 2. Defaults to pTrt1.
pCtrl2	The mean of the response in the control group for MRCT 2. Defaults to pTrt2.
randRatio1	The randomization ratio between the treatment group and control group for MRCT 1.
randRatio2	The randomization ratio between the treatment group and control group for MRCT 2. Defaults to randRatio1.
responseType	The type of response. One of "continuous" and "binary".
B	The number of simulation by Monto Carlo for responseType = "binary". Defaults to 100,000.

Details

The extended consistency probability via the extended riterion Method II of MHLW (2007) (conditional version),

$$\Pr\left(D_{k,\text{pool}} \geq 0, k = 1, \cdots, K \mid T^{(1)} > z_{1-\alpha}, T^{(2)} > z_{1-\alpha}\right) > 1 - \gamma,$$

is approximately

$$\frac{1}{(1-\beta_1)(1-\beta_2)} \int_{-z_{1-\beta_1}}^{\infty} \int_{-z_{1-\beta_2}}^{\infty} \Phi_K\left(\mathbf{0}; -\vec{\xi}(u,v), \Omega\right) \phi(u)\phi(v)\mathrm{d}u\mathrm{d}v,$$

where $\vec{\xi}$ and Ω are given in Qing et al (2025).

Since there is no closed form of above equation, conProbII2 utilizes the pmvnorm function for numerical integration.

For binary response, the above approximation loses precision under moderate sample size. Hence conProbII2 applies Monto Carlo to calculate the correct consistency probability.

The overall sample size is calculated in the same way as conProb.

Value

A list containing the following two components:

CP The consistency probability, a scalar.

N1 The overall sample size for MRCT 1.

N2 The overall sample size for MRCT 2.

Examples

```
# Example 3 in Qing et al (2025)
alpha <- 0.05
power1 <- 0.8
conProbII2(alpha = alpha, power1 = power1, rF1 = rep(1/2,2), d1 = 1,
sigmaTrt1 = 4, responseType = "continuous")
conProbII2(alpha = alpha, power1 = power1, rF1 = rep(1/3,3), d1 = 1,
sigmaTrt1 = 4, responseType = "continuous")
conProbII2(alpha = alpha, power1 = power1, rF1 = rep(1/4,4), d1 = 1,
sigmaTrt1 = 4, responseType = "continuous")
rFk1 <- 0.044
rFk2 <- rFk3 <- (1-rFk1)/2
rF <- c(rFk1, rFk2, rFk3)
conProbII2(alpha = alpha, power1 = power1, rF1 = rF, d1 = 1,
sigmaTrt1 = 4, responseType = "continuous")

# Example 4 in Qing et al (2025)
alpha <- 0.05
power1 <- 0.8
rFk1 <- 0.06
rFk2 <- rFk3 <- (1-rFk1)/2
rF <- c(rFk1, rFk2, rFk3)
conProbII2(alpha = alpha, power1 = power1, rF1 = rF, pTrt1 = 0.9,
pCtrl1 = 0.8, responseType = "binary")
rFk1 <- 0.044
rFk2 <- rFk3 <- (1-rFk1)/2
rF <- c(rFk1, rFk2, rFk3)
conProbII2(alpha = alpha, power1 = power1, rF1 = rF, pTrt1 = 0.9,
pCtrl1 = 0.8, responseType = "binary")
```

`CPlbd` *Lower bound of the consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model.*

Description

Calculate the lower bound of consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model for normal endpoint.

Usage

```
CPlbd(
  alpha,
  beta,
  tau_delta_r,
  delta,
  sigma0r,
  sigma1r,
  fr,
  pi = 0.5,
  randRatio = 1
)
```

Arguments

alpha	The one-sided Type I error.
beta	The Type II error.
tau_delta_r	The ratio of the standard deviation vs mean of the Gaussian prior for the regional treatment effect.
delta	The mean of the Gaussian prior for the regional treatment effect.
sigma0r	The standard deviation(s) of response in the treatment group for region(s) of interest. A scalar or vector of length equals to the total number of region(s) of interest.
sigma1r	The standard deviation(s) of response in the control group for region(s) of interest. A scalar or vector of length equals to sigma0r.
fr	The regional fraction(s) for the region(s) of interest. A scalar or vector of equal length to sigma0r and sigma1r.
pi	The threshold ratio in Method I of MHLW (2007)(conditional version). Defaults to 0.5.
randRatio	The randomization ratio between the treatment group and control group. Defaults to 1.

Details

The lower bound of the consistency probability via Method I of MHLW (2007)(conditional version) under random effects model, $\Pr\left(\widetilde{D}_r \geq \pi\widetilde{D}|T > z_{1-\alpha}\right)$, is approximately

$$\frac{1}{1-\beta}\int_{u>z_\beta}\Phi\left(\frac{2\delta^2(1-\pi)(u+z_{1-\alpha}+z_{1-\beta})}{\tau^2(z_{1-\alpha}+z_{1-\beta})^2}\right)d\Phi(u).$$

provided $\frac{\tau}{\delta} < \frac{\sqrt{2}}{z_{1-\alpha}+z_{1-\beta}}$. Otherwise, CP will also be greater than

$$\frac{1}{1-\beta}\int_{u>z_\beta}\Phi\left(\frac{(1-\pi)\,(u+z_{1-\alpha}+z_{1-\beta})}{\sqrt{\frac{\tau^2}{\delta^2}(z_{1-\alpha}+z_{1-\beta})^2-1}}\right)d\Phi(u).$$

. Since there is no closed form of above equation, CPlbd utilizes the integrate function for numerical integration.

The sample size(s) for the region(s) of interest at the lower bound are calculated based on solving the following equation for $n_0^{(r)}$,

$$\frac{h_r}{h_r+1} = \frac{\tau^2}{2\delta^2}(z_{1-\alpha}+z_{1-\beta})^2, \quad n_1^{(r)} = \ell n_0^{(r)}$$

where $h_r = \tau^2/\sigma^{2(r)}, \sigma^{2(r)} = \Omega_r/(n_0 f_r), \Omega_r = \ell^{-1}\sigma_1^{2(r)} + \sigma_1^{2(r)}$. Then $n^{(r)} = n_0^{(r)} + n_1^{(r)}$, where both of $n_0^{(r)}$ and $n_1^{(r)}$ should be integers and hence $n^{(r)}$.

Value

A list containing the following components:

CPlbd The lower bound of consistency probability, a scalar.

n0r The regional sample size of the control group. (n.a. when the condition for positive sample size is not met)

`n1r` The regional sample size of the treatment group. (n.a. when the condition for positive sample size is not met)

`nr` The regional sample size. (n.a. when the condition for positive sample size is not met)

Examples

```
CPlbd(alpha=0.05, beta=0.2, tau_delta_r=0.4, pi=0.5, delta=0.25,
      sigma0r=1, sigma1r=1, randRatio=1, fr=c(0.1))
CPlbd(alpha=0.05, beta=0.2, tau_delta_r=0.4, pi=0.5, delta=0.25,
      sigma0r=c(1,1), sigma1r=c(1,1), randRatio=1, fr=c(0.1,0.5))
```

---

regATE_dr *Doubly robust estimator for regional average treatment effects*

Description

This function implements a doubly robust estimator for the average treatment effect (ATE) in a specific region within a MRCT.

Usage

```
regATE_dr(Y, X, D, Trt, num_boot = 1000, conf_level = 0.95)
```

Arguments

`Y`	A numeric vector of the observed outcome.
`X`	A matrix or data frame of covariates.
`D`	A binary vector (0 or 1) indicating the region (1 for the target region).
`Trt`	A binary vector (0 or 1) indicating treatment assignment (1 for the treatment group).
`num_boot`	Integer. The number of bootstrap iterations for inference. Default is 1000.
`conf_level`	Numeric. Confidence level for the interval. Default is 1000.

Details

The estimator uses the formula:

$$\widehat{\tau}_{\mathrm{dr}} = \frac{1}{n \cdot \Pr(D=1)} \sum_{i=1}^{n} \Big[D_i\{\widehat{m}_1(X_i) - \widehat{m}_0(X_i)\} + \widehat{W}_{1,i}\{Y_i - \widehat{m}_1(X_i)\} - \widehat{W}_{0,i}\{Y_i - \widehat{m}_0(X_i)\} \Big]$$

where $\widehat{W}_{1,i}$ and $\widehat{W}_{0,i}$ are weights derived from the propensity scores of being in the target region and receiving treatment.

Value

A list containing:

`estimate`	The point estimate of the regional ATE.
`se`	The bootstrap standard error.
`conf_int`	A named vector with the lower and upper confidence limits.
`p_value_two_sided`	The p-value for a two-sided test (H_0 : regional ATE $= 0$).
`p_value_one_sided`	The p-value for a one-sided test (H_0 : regional ATE ≤ 0).

Examples

```
set.seed(42)
sampleSize <- 300
regionProp <- 0.16
sampleSizeRegion <- round(regionProp * sampleSize)
sampleSizeNon    <- sampleSize - sampleSizeRegion
# define region and treatment (1:1 randomization)
region <- c(rep(1, sampleSizeRegion), rep(0, sampleSizeNon))
trt <- c(rep(1, sampleSizeRegion/2), rep(0, sampleSizeRegion/2),
         rep(1, sampleSizeNon/2),    rep(0, sampleSizeNon/2))
# generate covariates with regional heterogeneity
x1 <- c(rnorm(sampleSizeRegion, mean = 0.5), rnorm(sampleSizeNon, mean = 1.0))
x2 <- rnorm(sampleSize, mean = 0)
X  <- data.frame(x1 = x1, x2 = x2)
# generate outcome
epsilon <- rnorm(sampleSize, mean = 0)
Y <- x1 + x2 + trt * (x1 + x2) + epsilon
# estimation regATE_dr
results <- regATE_dr(Y = Y, X = X, D = region, Trt = trt, num_boot = 1000)
print(results)
```

---

regFrac *Regional fraction for MRCT via criterion Method I of MHLW (2007) (conditional version) under fixed effects model*

Description

Calculate the minimal regional fraction given the consistency probability for MRCT via criterion Method I of MHLW (2007) (conditional version) under fixed effects model.

Usage

```
regFrac(
  alpha,
  power,
  pi = 0.5,
  CP = 0.8,
  d = 1,
  sigmaTrt = 4,
  sigmaCtrl = sigmaTrt,
  randRatio = 1
)
```

Arguments

alpha	The Type I error.
power	Power.
pi	The threshold ratio in criterion Method I of MHLW (2007) (conditional version). Defaults to 0.5.
CP	the consistency probability. Defaults to 80%.
d	The true mean of difference of response. Defaults to 1.
sigmaTrt	The standard deviation of response in the treatment group. Defaults to 4.
sigmaCtrl	The standard deviation of response in the control group. Defaults to sigmaTrt.
randRatio	The randomization ratio between the treatment group and control group. Defaults to 1.

Details

Given the consistency probability, there is a minimal regional fraction rF. To calculate the minimal rF, regFrac utilizes two core computational components:

- The conProb function to compute the (conditional) consistency probability
- The uniroot function from the **stats** package for numerical root-finding

The solution is obtained by solving the following equation numerically:

$$\texttt{conProb(rF)} - \texttt{CP} = 0$$
$$\text{where } \texttt{rF} \in (0, 1).$$

The overall sample size is obtain by conProb simultaneously.

Value

A list containing the following two components:

rF The regional fraction, a scalar.

N The overall sample size.

Examples

```
conProb(alpha = 0.05, power = 0.8, rF = 0.271, d = 1, sigmaTrt = 4)

regFrac(alpha = 0.05, power = 0.8, d = 1, sigmaTrt = 4)
```

`regFrac2` *Regional fraction for two MRCTs via the extended criterion Method I of MHLW (2007) (conditional version) under fixed effects model*

Description

Calculate the optimal regional fractions given the consistency probability for two MRCTs via the extended criterion Method I of MHLW (2007) (conditional version) under fixed effects model

Usage

```
regFrac2(alpha, power, pi = 0.5, CP = 0.8)
```

Arguments

`alpha`	The Type I error.
`power`	Power for MRCT 1 and 2.
`pi`	The threshold ratio in the extended criterion Method I of MHLW (2007) (conditional version). Defaults to 0.5.
`CP`	The consistency probability. Defaults to 80%.

Details

If we assume equal treatment effects, common regional fractions, common powers and common randomization ratios across two studies, i.e., $\sigma^{2(\mathrm{t},1)} = \sigma^{2(\mathrm{t},2)}, \sigma^{2(\mathrm{c},1)} = \sigma^{2(\mathrm{c},2)}, d^{(1)} = d^{(2)}, f_k^{(1)} = f_k^{(2)} = f_k, \beta_1 = \beta_2$ and $r^{(1)} = r^{(2)}$, then the CP reduces to

$$\frac{1}{(1-\beta)^2}\int_{-z_{1-\beta}}^{\infty}\int_{-z_{1-\beta}}^{\infty}\Phi\left(\frac{(1-\pi)(2z_{1-\alpha}+2z_{1-\beta}+u+v)}{\sqrt{2(f_k^{-1}-1)}}\right)\phi(u)\phi(v)\mathrm{d}u\mathrm{d}v.$$

Given the consistency probability, there is a minimal regional fraction `rF`.

The overall sample size is obtain by `conProb2` simultaneously.

Value

A list containing the following one component:

`rF` The regional fraction for MRCT 1 and 2, a scalar.

Examples

```
regFrac2(alpha = 0.05, power = 0.8)
```

`ssCPbinary` *Overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model.*

Description

Calculate the overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model for the binary endpoint.

Usage

```
ssCPbinary(alpha, beta, p0r, p1r, pi = 0.5, randRatio = 1, target, f)
```

Arguments

`alpha`	Type I error for one-sided test.
`beta`	Type II error.
`p0r`	vector of probabilities in the control group. a `R x 1` vector with each component representing the probability in region r.
`p1r`	vector of probabilities in the treatment group. a `R x 1` vector with each component representing the probability in region r.
`pi`	threshold ratio in Method I of MHLW (2007)(conditional version). Defaults to 0.5.
`randRatio`	randomization ratio between the treatment group and control group. Defaults to 1.
`target`	index for the region of interest. an integer of value referring to the region with the corresponding position in `f`.
`f`	vector of regional fractions.a `R x 1` vector with each component representing the fraction in region r and the sum must equal 1.

Details

The overall sample size is calculated based on the following equation,

$$\sum_r \left(\tau^2 + \frac{\Omega_r}{n_0 f_r}\right)^{-1} = \frac{(z_{1-\alpha} + z_{1-\beta})^2}{\delta^2} \quad n_1 = \ell n_0.$$

Since there is no closed form of above equation, `ssCPbinary` utilizes the `uniroot` function for numerical solution for n_0. Then $n = n_0 + n_1$.

The consistency probability via Method I of MHLW (2007)(conditional version) under random effects model, $\Pr\left(\widetilde{D}_r \geq \pi \widetilde{D} | T > z_{1-\alpha}\right)$, is approximately

$$\frac{1}{1-\beta} \int_{u > z_\beta} \Phi\left(\frac{(1-\pi)(u + z_{1-\alpha} + z_{1-\beta})}{\sqrt{\rho_r^{-1} - 1}}\right) d\Phi(u)$$

where $\rho_r^{-1} - 1 = \frac{h_r}{h_r+1} \sum_{j \neq r}^{R} \frac{h_j}{h_j+1}, w = \frac{1}{\tau^2} \sum_{j=1}^{R} \frac{h_j}{h_j+1}, h_r = \tau^2/\sigma^{2(r)}, \sigma^{2(r)} = \Omega_r/(n_0 f_r)$. Since there is no closed form of above equation, `ssCPbinary` utilizes the `integrate` function for numerical integration.

For the binary endpoint, $\Omega_r = \ell^{-1}\sigma_1^{2(r)} + \sigma_1^{2(r)}, \sigma_1^{2(r)} = p_1^{(r)}(1-p_1^{(r)}), \sigma_0^{2(r)} = p_0^{(r)}(1-p_0^{(r)})$. δ and τ are derived based on the naive approach as the mean and standard deviation of the regional risk differences (i.e., $p_1^{(r)} - p_0^{(r)}$).

Value

A list containing the following components:

`n0` The overall sample size of the control group

`n1` The overall sample size of the treatment group

`n` The overall sample size.

`CP` The consistency probability, a scalar.

`"error: condition not met. negative n0 returned"`

`"error: regional fractions do not sum up to 1"` When vector f does not sum up to 1

Examples

```
ssCPbinary(alpha=0.025, beta=0.2, p0r=rep(0.3,3),
           p1r=c(0.6,0.4,0.2)+rep(0.3,3), pi=0.5,
           randRatio=1, target=1, f=rep(1/3,3))
```

`ssCPnorm` *Overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model.*

Description

Calculate the overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model for the normal endpoint.

Usage

```
ssCPnorm(
  alpha,
  beta,
  delta,
  tau,
  sigma1r,
  sigma0r,
  pi = 0.5,
  randRatio = 1,
  target,
  f
)
```

Arguments

alpha	Type I error for one-sided test.
beta	Type II error.
delta	mean of the Gaussian prior for the regional treatment effect.
tau	standard deviation of the Gaussian prior for the regional treatment effect.
sigma1r	regional standard deviations of the response in the treatment group, a R x 1 vector with each component representing the standard deviation in region r .
sigma0r	regional standard deviations of the response in the control group. a R x 1 vector with each component representing the standard deviation in region r.
pi	threshold ratio in Method I of MHLW (2007)(conditional version). Defaults to 0.5.
randRatio	randomization ratio between the treatment group and control group. Defaults to 1.
target	The index for the region of interest. an integer of value referring to the region with the corresponding position in f.
f	regional fractions, a R x 1 vector with each component representing the fraction in region r and the sum must equal 1.

Details

The overall sample size is calculated based on the following equation,

$$\sum_r \left(\tau^2 + \frac{\Omega_r}{n_0 f_r}\right)^{-1} = \frac{(z_{1-\alpha} + z_{1-\beta})^2}{\delta^2} \quad n_1 = \ell n_0.$$

Since there is no closed form of above equation, ssCPnorm utilizes the uniroot function for numerical solution for n_0. Then $n = n_0 + n_1$.

The consistency probability via Method I of MHLW (2007)(conditional version) under random effects model, $\Pr\left(\widetilde{D}_r \geq \pi\widetilde{D}|T > z_{1-\alpha}\right)$, is approximately

$$\frac{1}{1-\beta}\int_{u>z_\beta} \Phi\left(\frac{(1-\pi)(u + z_{1-\alpha} + z_{1-\beta})}{\sqrt{\rho_r^{-1} - 1}}\right) d\Phi(u)$$

where $\rho_r^{-1} - 1 = \frac{h_r}{h_r+1}\sum_{j\neq r}^{R}\frac{h_j}{h_j+1}, w = \frac{1}{\tau^2}\sum_{j=1}^{R}\frac{h_j}{h_j+1}, h_r = \tau^2/\sigma^{2(r)}, \sigma^{2(r)} = \Omega_r/(n_0 f_r)$. Since there is no closed form of above equation, ssCPnorm utilizes the integrate function for numerical integration.

For the normal endpoint, $\Omega_r = \ell^{-1}\sigma_1^{2(r)} + \sigma_1^{2(r)}$

Value

A list containing the following components:

n0 The overall sample size of the control group

n1 The overall sample size of the treatment group

n The overall sample size.

CP The consistency probability, a scalar.

"error: condition not met. negative n0 returned"

"error: regional fractions do not sum up to 1" When vector f does not sum up to 1

Examples

```
ssCPnorm(alpha=0.025, beta=0.1, delta=0.25, tau=0.1,
         sigma1r=rep(1,3),sigma0r=rep(1,3),
         randRatio=1, pi=0.5, target=1, f=c(1/3,1/3,1/3))
```

`ssCPsurvNPH` *Overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model.*

Description

Calculate the overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model for the survival endpoint with nPH assumption.

Usage

```
ssCPsurvNPH(
  alpha,
  beta,
  lambda0a,
  lambda0b,
  lambda1a,
  lambda1b,
  breakpoint,
  shape0,
  scale0,
  shape1,
  scale1,
  PW = T,
  unimin,
  unimax,
  pi = 0.5,
  randRatio = 1,
  target,
  f,
  eta
)
```

Arguments

`alpha`	Type I error for one-sided test.
`beta`	Type II error.
`lambda0a`	vector rate parameter of the piece-wise exponential distribution on or before the breakpoint in the control group. a `R x 1` vector with each component representing the rate parameter in region r.
`lambda0b`	vector rate parameter of the piece-wise exponential distribution after the breakpoint in the control group. a `R x 1` vector with each component representing the rate parameter in region r.

lambda1a	vector rate parameter of the piece-wise exponential distribution on or before the breakpoint in the treatment group. a R x 1 vector with each component representing the rate parameter in region r.
lambda1b	vector rate parameter of the piece-wise exponential distribution before the breakpoint in the treatment group. a R x 1 vector with each component representing the rate parameter in region r.
breakpoint	scalar of the time at which the hazard changes. Assumed universal breakpoint value across groups and regions.
shape0	vector of the shape parameter of the weibull distribution in the control group (shape=1 reduces to exponential). a R x 1 vector with each component representing the shape parameter in region r.
scale0	vector of the scale parameter of the weibull distribution in the control group (scale=1/rate). a R x 1 vector with each component representing the scale parameter in region r.
shape1	vector of the shape parameter of the weibull distribution in the treatment group(shape=1 reduces to exponential). a R x 1 vector with each component representing the shape parameter in region r.
scale1	vector of the scale parameter of the weibull distribution in the treatment group (scale=1/rate). a R x 1 vector with each component representing the scale parameter in region r.
PW	logical; if TRUE (default), survival time follow piece-wise exponential distribution. Otherwise Weibull distribution
unimin	vector of the lower limits of the uniform distribution. a R x 1 vector with each component representing the lower limit in region r.
unimax	vector of the upper limits of the uniform distribution. a R x 1 vector with each component representing the upper parameter in region r.
pi	threshold ratio in Method I of MHLW (2007)(conditional version). Defaults to 0.5.
randRatio	randomization ratio between the treatment group and control group. Defaults to 1.
target	index for the region of interest. an integer of value referring to the region with the corresponding position in f.
f	vector of regional fractions. a R x 1 vector with each component representing the regional fraction in region r and the sum must equal 1.
eta	a scalar of truncated time in RMST. Assumed universal truncated time across groups and regions.

Details

Assume the survival endpoint follows either piece-wise exponential distribution with hazard function (i.e., rate function) $\lambda_k^{(r)} I(0 < t \leq \psi) + \gamma_k^{(r)} I(t > \psi)$, where ψ(breakpoint) is the change point and $\lambda_k^{(r)}$ (lambda0a, lambda0b), $\gamma_k^{(r)}$ (lambda1a, lambda1b) represent the hazard on or before and after the change point ψ in group k of region r. or Weibull distribution (i.e., Weibull($\nu_k^{(r)}, \theta_k^{(r)}$)), with shape parameter $\nu_k^{(r)}$ (shape0,shape1) and scale parameter (i.e., 1/rate parameter) $\theta_k^{(r)}$ (scale0, scale1) for group k region r. Assume the censoring time follows uniform distribution with lower limit unimin and upper limit unimax.

The overall sample size is calculated based on the following equation,

$$\sum_r \left(\tau^2 + \frac{\Omega_r}{n_0 f_r}\right)^{-1} = \frac{(z_{1-\alpha} + z_{1-\beta})^2}{\delta^2} \quad n_1 = \ell n_0.$$

Since there is no closed form of above equation, ssCPsurvPH utilizes the uniroot function for numerical solution for n_0. Then $n = n_0 + n_1$.

The consistency probability via Method I of MHLW (2007)(conditional version) under random effects model, $\Pr\left(\widetilde{D}_r \geq \pi\widetilde{D} | T > z_{1-\alpha}\right)$, is approximately

$$\frac{1}{1-\beta}\int_{u>z_\beta} \Phi\left(\frac{(1-\pi)(u + z_{1-\alpha} + z_{1-\beta})}{\sqrt{\rho_r^{-1} - 1}}\right) d\Phi(u)$$

where $\rho_r^{-1} - 1 = \frac{h_r}{h_r+1}\sum_{j\neq r}^{R}\frac{h_j}{h_j+1}, w = \frac{1}{\tau^2}\sum_{j=1}^{R}\frac{h_j}{h_j+1}, h_r = \tau^2/\sigma^{2(r)}, \sigma^{2(r)} = \Omega_r/(n_0 f_r)$. Since there is no closed form of above equation, ssCPsurvPH utilizes the integrate function for numerical integration.

For survival endpoint under non-PH assumption, $\Omega_r = \ell^{-1}\sigma_1^{2(r)} + \sigma_1^{2(r)}$. δ and τ are derived based on the naive approach as the mean and standard deviation of the regional RMST between group difference.

Value

A list containing the following components:

`Dr` regional effect

`sigma2.0r` asymptotic vairance of RMST estimate of the control group in region r

`sigma2.1r` asymptotic vairance of RMST estimate of the treatment group in region r

`n0` The overall sample size of the control group

`n1` The overall sample size of the treatment group

`n` The overall sample size.

`CP` The consistency probability, a scalar.

`"error: condition not met. negative n0 returned"`

`"error: regional fractions do not sum up to 1"` When vector f does not sum up to 1

Examples

```
ssCPsurvNPH(alpha=0.025, beta=0.2,
            lambda0a=c(0.07,0.07,0.07,0.07),lambda0b=c(0.03,0.04,0.05,0.06),
            lambda1a=c(0.02,0.03,0.04,0.05),lambda1b=c(0.03,0.04,0.05,0.06),
            breakpoint=10,
            unimin=rep(0,4),unimax=rep(80*3,4),
            pi=0.5, randRatio=1, target=1, f=rep(1/4,4), eta=80)
```

ssCPsurvPH *Overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model.*

Description

Calculate the overall sample size and consistency probability for one MRCT via Method I of MHLW (2007)(conditional version) under random effects model for survival endpoint with PH assumption.

Usage

```
ssCPsurvPH(alpha, beta, lambda0r, hr, pi = 0.5, randRatio = 1, target, f, L)
```

Arguments

alpha	one-sided Type I error.
beta	Type II error.
lambda0r	vector of rate parameters in the control group. a R x 1 vector with each component representing the rate parameter in region r.
hr	vector of hazard ratios of treatment group vs control group. a R x 1 vector with each component representing the hr in region r.
pi	threshold ratio in Method I of MHLW (2007)(conditional version). Defaults to 0.5.
randRatio	randomization ratio between the treatment group and control group. Defaults to 1.
target	index for the region of interest. an integer of value referring to the region with the corresponding position in f.
f	vector of regional fractions. a R x 1 vector with each component representing the regional fraction in region r and the sum must equal 1.
L	scalar of follow-up time in a study assuming fixed study duration design.

Details

Assume the survival endpoint in region r of group k follows exponential distribution with rate parameter $\lambda_k^{(r)}$: $T_k^{(r)} \sim \exp(\lambda_k^{(r)}), k = 0, 1, r = 1 \ldots R$. Assume the censoring time follows uniform distribution with lower limit unimin and upper limit unimax. Assume a study with fixed study duration where each subject is followed for the same length in time (i.e.,L time units) and administrative censoring at the end of study follow-up is the only type of censoring.

The overall sample size is calculated based on the following equation,

$$\sum_r \left(\tau^2 + \frac{\Omega_r}{n_0 f_r}\right)^{-1} = \frac{(z_{1-\alpha} + z_{1-\beta})^2}{\delta^2} \quad n_1 = \ell n_0.$$

Since there is no closed form of above equation, ssCPsurvPH utilizes the uniroot function for numerical solution for n_0. Then $n = n_0 + n_1$.

The consistency probability via Method I of MHLW (2007)(conditional version) under random effects model, $\Pr\left(\widetilde{D}_r \geq \pi\widetilde{D}|T > z_{1-\alpha}\right)$, is approximately

$$\frac{1}{1-\beta}\int_{u>z_\beta}\Phi\left(\frac{(1-\pi)(u+z_{1-\alpha}+z_{1-\beta})}{\sqrt{\rho_r^{-1}-1}}\right)d\Phi(u)$$

where $\rho_r^{-1}-1=\frac{h_r}{h_r+1}\sum_{j\neq r}^{R}\frac{h_j}{h_j+1}, w=\frac{1}{\tau^2}\sum_{j=1}^{R}\frac{h_j}{h_j+1}, h_r=\tau^2/\sigma^{2(r)}, \sigma^{2(r)}=\Omega_r/(n_0 f_r)$. Since there is no closed form of above equation, `ssCPsurvPH` utilizes the `integrate` function for numerical integration.

Note for survival endpoint under PH assumption, $\Omega_r=(\ell+1)^2/\{\ell(P_0^{(r)}+\ell P_1^{(r)})\}, P_k^{(r)}=1-e^{-\lambda_k^{(r)}L}$. δ and τ are derived based on the naive approach as the mean and standard deviation of the regional hazard ratios.

Value

A list containing the following components:

`n0` The overall sample size of the control group

`n1` The overall sample size of the treatment group

`n` The overall sample size.

`CP` The consistency probability, a scalar.

`"error: condition not met. negative n0 returned"`

`"error: regional fractions do not sum up to 1"` When vector f does not sum up to 1

Examples

```
ssCPsurvPH(alpha=0.025, beta=0.2, lambda0r=rep(0.05,3), hr=c(0.7,0.6,0.4),
           pi=0.5, randRatio=1, target=1, f=rep(1/3,3), L=36)
```

`table2MRCTs` *Reproduce all tables in Qing et al (2025)*

Description

This function reproduces the simulation results for tables 1 through 6 from Qing et al (2025).

Usage

```
table2MRCTs(numSimu, choice, seed = 1)
```

Arguments

`numSimu`	Integer. The number of Monte Carlo simulations to run (e.g., 10000).
`choice`	Integer (1-6). Specifies which table to generate.
`seed`	Integer. Random seed for reproducibility. Default is 1.

Details

- Tables 1 & 2: Single MRCT with Binary and Continuous outcomes.
- Tables 3 & 4: Two MRCTs with Binary and Continuous outcomes.
- Tables 5 & 6: Sensitivity analysis with different randomization ratios and powers.

Value

A data frame containing the simulation parameters, sample sizes, theoretical region fractions, and empirical consistency probabilities.

Examples

```
## Not run:
for (choice in 1:6) {
print(paste("Table", choice))
print(table2MRCTs(100000,choice,1))
}
## End(Not run)
```

Index

conProb, *2*, 3, *5*, *7*, *9*, *14*
conProb2, *2*, 4, *15*
conProbII, 6
conProbII2, 8
CPlbd, 10

integrate, *3*, *5*, *11*, *16*, *18*, *21*, *23*

pmvnorm, *7*, *9*

regATE_dr, 12
regFrac, *2*, 13
regFrac2, *2*, 15

ssCPbinary, *2*, 16
ssCPnorm, *2*, 17
ssCPsurvNPH, *2*, 19
ssCPsurvPH, *2*, 22
ssMRCT-package, 2

table2MRCTs, 23

uniroot, *14*, *16*, *18*, *21*, *22*